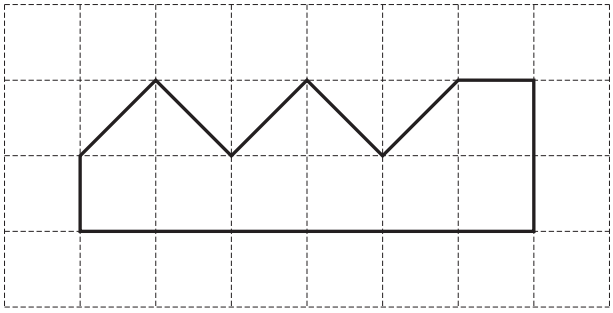
0580_s23_qp_33_Q3

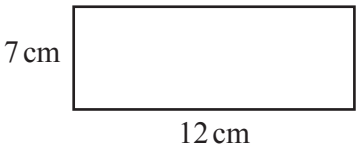
3 **(a)** The diagram shows a shape on a 1 cm^2 grid.



Work out the area of the shape.

..... cm^2 [1]

(b)



NOT TO SCALE

Work out the perimeter of the rectangle.

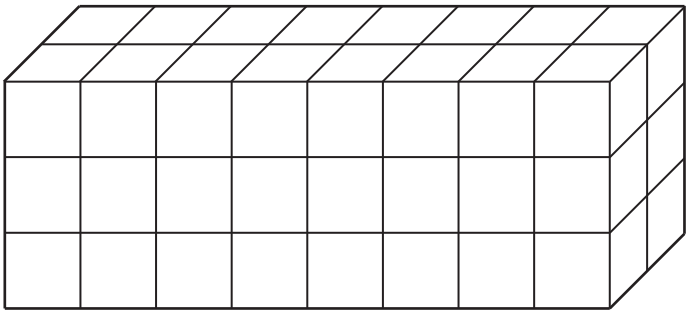
..... cm [1]

(c) A square has an area of 841 cm^2.

Work out the length of one side of the square.

..... cm [1]

(d) The diagram shows a cuboid made from 1 cm^3 cubes.



NOT TO SCALE

(i) Work out the volume of the cuboid.

..... cm^3 [1]

(ii) Write down the dimensions of a different cuboid that can be made using all of the cubes.

..... cm by ..... cm by ..... cm [1]

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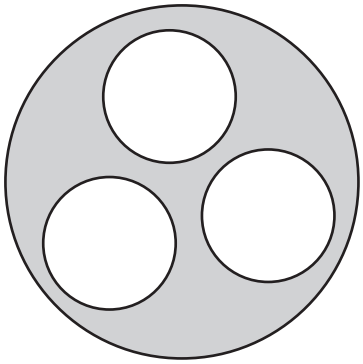
(e)



NOT TO SCALE

The diagram shows three small circles and one large circle.
The large circle has radius 20 cm.
The small circles each have radius 4 cm.

Work out the shaded area.
Give your answer in terms of π.

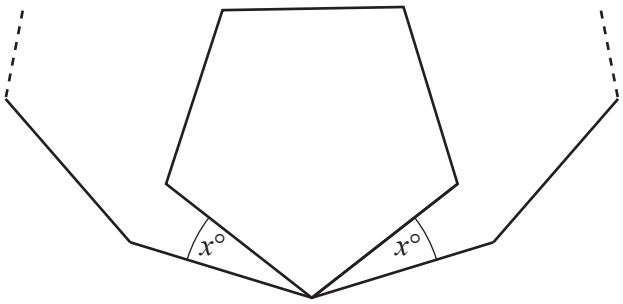
..... cm^2 [3]

(f) The exterior angle of a 9-sided regular polygon is 40°.

(i) Work out the size of the interior angle of this polygon.

..... [1]

(ii)



NOT TO SCALE

The diagram shows a regular pentagon inside part of a regular 9-sided polygon.

Work out the value of x.

x = ... [4]